Portfolio Optimization Primer

Financial Sector | 2/11/2026

Overview of Multifactor Model and Hierarchical Risk-Parity

Multifactor Model

Overview of Optimization Techniques

The multifactor model (MF) identifies which characteristics most consistently drive returns in the financial sector by analyzing historical data across six factors: TTM momentum, price-to-book, earnings yield, ROE, debt-to-equity, and beta. These factors generate expected returns that feed into an optimization process designed to maximize performance while controlling risk. The model is well suited to financial stocks because it captures the sector's core drivers: value, quality, risk, and trend. While systematic and evidence-based, it assumes historical relationships persist and may still produce concentrated portfolios.

Economic Interpretation

Because of its structural design, the factored portfolio reflects sensitivities to factor exposures rather than broad diversification across all holdings. In this financial-sector portfolio, the factor optimizer increases exposure to AMP (+7.62%), BX (+7.54%), MKL (+5.07%), MS (+3.32%), and V (+3.95%), while fully eliminating O (-15.72%) and PLD (-11.77%). These shifts are driven by the selected value, quality, momentum, and risk characteristics rather than macroeconomic forecasts or discretionary views. By concentrating capital in stocks that score most favorably across the factor framework, the model increases alignment with systematic return drivers, though it also introduces greater concentration risk and sensitivity to changes in factor performance.

Hierarchical Risk-Parity (HRP)

Overview of Optimization Techniques

In the financial sector, where equities often move together due to shared exposure to interest rates, credit cycles, and macroeconomic conditions, HRP helps limit concentration risk by allocating capital across clusters of related firms rather than individual securities. By grouping financial stocks such as banks, insurers, and asset managers based on their historical correlation structure, HRP improves diversification relative to traditional Risk Parity while reducing reliance on precise pairwise correlation estimates. However, portfolio outcomes still depend on the chosen clustering methodology, and different grouping choices can lead to meaningfully different allocations.

Economic Interpretation

By clustering stocks based on their co-movement, HRP limits risk concentration across highly correlated financial subsectors, leading to more balanced exposure across banks, asset managers, insurers, and diversified financials. In this portfolio, HRP meaningfully increases allocations to MKL (+20.68%) and V (+14.41%), while modestly increasing exposure to O (+0.05%). At the same time, it reduces weights in AMP (-8.18%), BX (-8.34%), MS (-13.09%), and PLD (-4.53%), reflecting their stronger correlation within shared rate- and credit-sensitive clusters. These adjustments are driven by correlation structure rather than fundamental views, lowering concentration in closely related financial names. Because HRP dynamically reallocates based on evolving correlation patterns rather than fixed volatility assumptions, it promotes a more resilient mix of cyclical and diversified financial exposures across changing market conditions.

Portfolio Optimization Primer

Financial Sector | 2/11/2026

Overview of Mean Variance and Black-Litterman

Mean Variance

Overview of Optimization Techniques

Mean-Variance optimization chooses portfolio weights by balancing two goals: higher returns and smaller ups and downs. Rather than weighing everything equally, it accounts for how each stock behaves and how the stocks move together, then favors combinations that have offered a better tradeoff between reward and stability. Its strengths are that it follows a clear, widely used framework, encourages diversification by considering relationships between holdings, and provides a structured way to be more aggressive or conservative. Its weaknesses are that small input changes can lead to different results, it can concentrate too much on a few names without limits, and it can struggle when market conditions shift.

Economic Interpretation

Because of its heavy reliance on historical return data, Mean-Variance Optimization exhibits a strong bias toward assets that have thrived in the recent high-interest-rate environment while aggressively penalizing those that lagged. In this financials portfolio, the optimizer completely abandons interest-rate sensitive Real Estate Investment Trusts, cutting allocations to Realty Income (-15.72%) and Prologis (-11.76%) entirely to zero. Instead, it concentrates capital into historically strong performers like Visa (+3.95%), Blackstone (+7.54%), and Morgan Stanley (+3.32%), pushing them all to the 20% position limit. These shifts are driven by the backward-looking assumption that the trends of the past cycle will continue indefinitely. Mean-Variance works best when market conditions remain stable, but it fails here to account for the potential reversal in monetary policy that would likely favor the very yield-sensitive assets it has excluded.

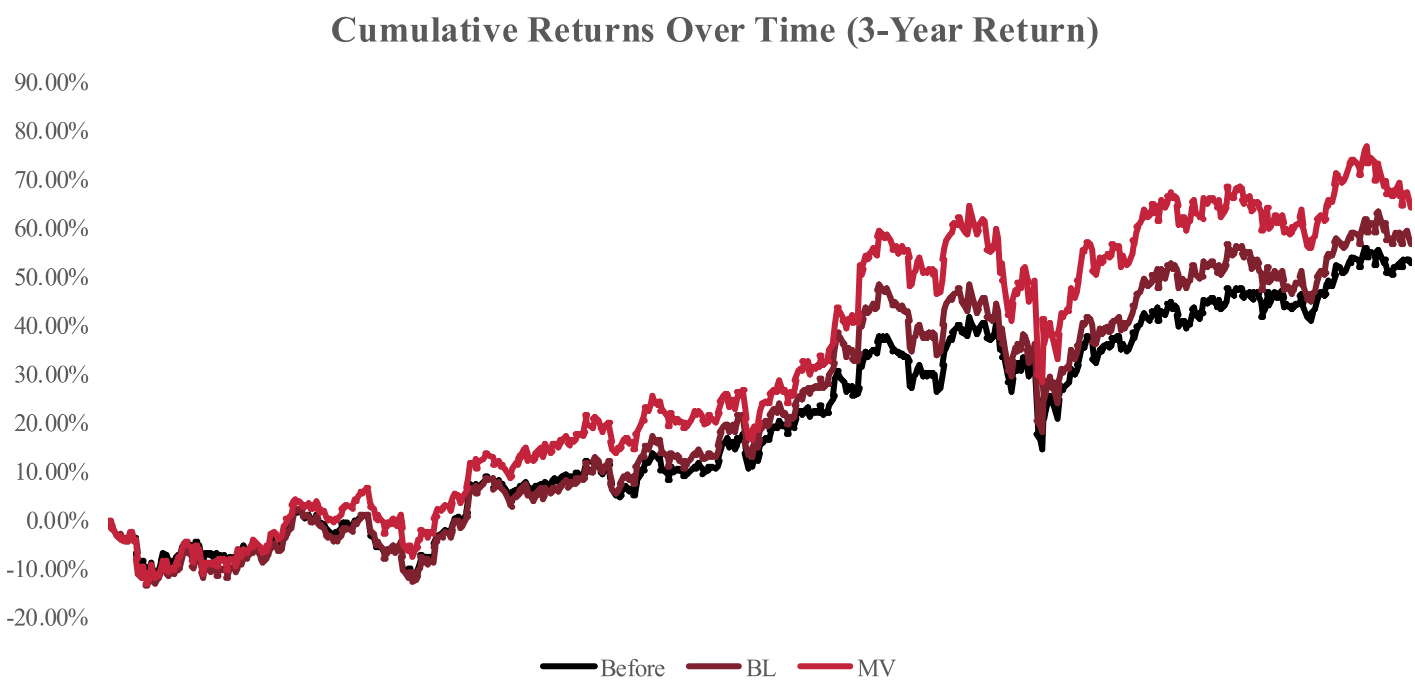
Black-Litterman

Overview of Optimization Techniques

Sentiment-Enhanced Black–Litterman builds on the standard Black–Litterman approach by combining two sources of "what we think will happen next": the team's own views and the tone of recent financial news. In practice, it reads current headlines about each company, summarizes whether the coverage is mostly positive or negative, and then uses that information to gently tilt the portfolio toward stocks with stronger sentiment and away from those with weaker sentiment, while still keeping the overall portfolio diversified. The benefit is that it makes the process less dependent on one person's opinion and more responsive to new information that may be emerging in the market. The tradeoff is that it only works well if the news is reliable and consistently available, and it adds extra moving parts that can amplify noise when headlines are misleading, overly hyped, or intentionally manipulated.

Economic Interpretation

Because of its ability to integrate forward-looking macroeconomic views, Black-Litterman exhibits sensitivities to the anticipated shift in Federal Reserve interest rate policy rather than just past price action. In this financials portfolio, the Black-Litterman optimizer aggressively rotates capital into rate-sensitive income plays, increasing allocations to Realty Income (+4.28%) to reach the 20% position limit, effectively reversing the "sell" signal generated by standard models. Conversely, it dramatically reduces exposure to Visa (-16.05%), exiting the position entirely to redeploy capital into higher-beta financial engines like Blackstone (+7.54%) and Morgan Stanley (+3.32%). These shifts are driven by the specific confidence levels assigned to interest rate forecasts, which prioritize assets with high "Rate Beta" over defensive compounders. Black-Litterman works best here by pricing in the benefits of a falling cost of capital, allowing the portfolio to proactively position for a soft landing rather than reactively defending against the last recession.

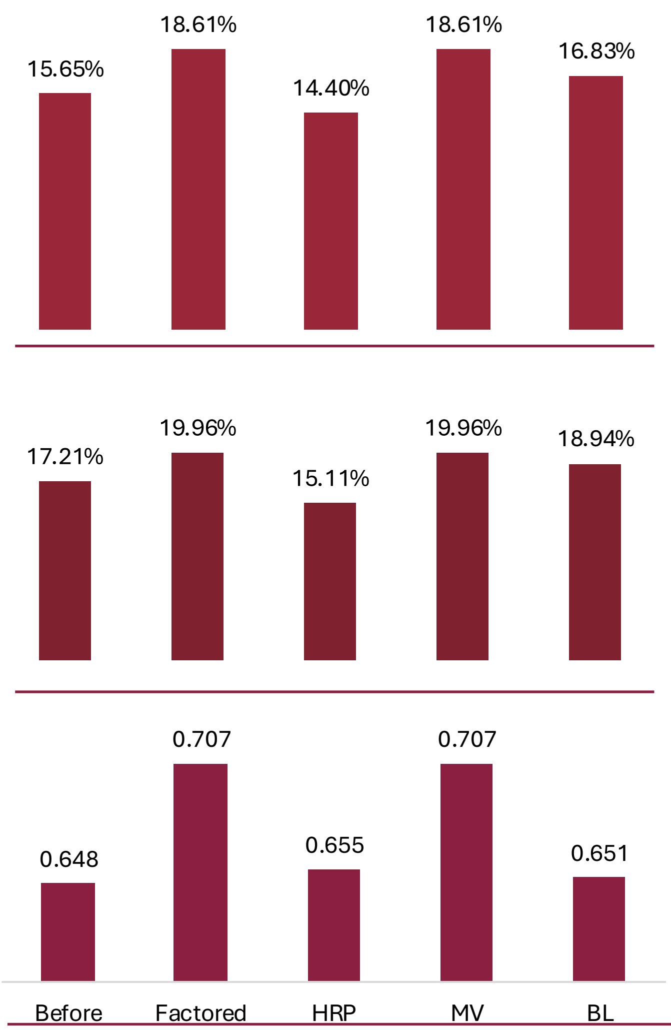


Portfolio Optimization Primer

Financial Sector | 2/11/2026

Comparative Analysis

Optimizer	Annual Return	Annual Vol	Sharpe Ratio	Sortino Ratio
Before	**15.65%**	**17.21%**	**0.648**	**0.936**
Factored	**18.61%**	**19.96%**	**0.707**	**1.024**
HRP	**14.40%**	**15.11%**	**0.655**	**0.963**
MV	**18.61%**	**19.96%**	**0.707**	**1.024**
BL	**16.83%**	**18.94%**	**0.651**	**0.940**



Expected return (Annual Return)

Highest (MV/Factored): 18.61%
Lowest (HRP): 14.40%

Interpretation: The Mean-Variance and Factored models achieved the highest returns (18.61%) by aggressively concentrating 100% of capital into high-beta asset managers and banks (BX, MS, AMP). In contrast, HRP produced the lowest return (14.40%) because its mandate for diversification forced it to allocate significant capital to lower-beta defensive assets like Visa, sacrificing raw upside for balance.

Volatility (Annual Vol)

Lowest (HRP): 15.11%
Highest (MV/Factored): 19.96%

Interpretation: Hierarchical Risk Parity (HRP) achieved the lowest volatility (15.11%) by correctly identifying Visa and Markel as "uncorrelated" diversifiers and overweighting them to dampen swings. The Mean-Variance model exhibited the highest volatility (19.96%) because it ignored these defensive buffers entirely, doubling down on highly correlated cyclical assets to chase maximum growth.

Sharpe ratio (Risk Adj)

Highest (MV/Factored): 0.707
Lowest (BL): 0.651

Interpretation: The Mean-Variance model posted the highest Sharpe Ratio (0.707) by "overfitting" to the recent bull run in capital markets stocks, assuming their high efficiency will repeat. The Black-Litterman model produced a lower ratio (0.651) because it intentionally deviated from this historical efficiency to position for a future interest rate regime (buying REITs), accepting lower backtested stats today for economic resilience tomorrow.

Analysis:

After analyzing the optimization models, Markel (MKL) and Morgan Stanley (MS) demonstrated the strongest capital retention, with MKL receiving a dominant 35.61% allocation in HRP and consistent 20% caps across all other models. Prologis (PLD) had the lowest average, being completely excluded (0.00%) from the Factored, Mean-Variance, and Black-Litterman models due to its high interest rate sensitivity and weaker historical efficiency. Visa (V) showed significant variability, receiving a massive 30.46% allocation in HRP as a diversifier, but complete exclusion from Black-Litterman (0.00%), indicating that while it offers statistical safety, forward-looking macro views prioritized higher-beta assets like Blackstone.

Capital Allocation For Each Stock

Tickers	Before	Factored	HRP	MV	BL
AMP	12.38%	20.00%	4.20%	20%	20%
BX	12.46%	20.00%	4.12%	20%	20%
MKL	14.93%	20.00%	35.61%	20%	20%
MS	16.68%	20.00%	3.59%	20%	20.0%
O	15.72%	0.00%	14.77%	0%	20%
PLD	11.76%	0.00%	7.23%	0%	0%
V	16.05%	20.00%	30.46%	20%	0%

Recommended Optimization Framework: Average of All the Methods

A blended optimization framework is applied to the financial sector portfolio by averaging allocations from the Factored, HRP, Mean-Variance, and Black–Litterman models, creating a balanced and resilient weighting scheme. This approach avoids overreliance on any single methodology, reducing sensitivity to estimation error, extreme assumptions, or model-specific biases that can distort portfolio construction. By combining multiple perspectives on risk, return, and correlation, the resulting allocations smooth out concentration effects while maintaining exposure to the sector's core drivers. The averaged weights reflect a practical compromise between return-seeking and risk control, producing a portfolio that is less reactive to short-term noise and more stable across changing interest-rate, credit, and macroeconomic environments than any individual optimization method alone.

Fit with the Nature of Financial Businesses

This blended framework fits financial businesses particularly well because the sector is heavily influenced by macro-driven risks, including interest-rate shifts (MS, AMP), credit cycle exposure (BX), underwriting and balance-sheet strength (MKL), and transaction or payment volume sensitivity (V). This is reflected in the allocation table, where the averaged approach maintains meaningful exposure to diversified financial leaders like MKL (23.90%) and V (17.62%), while keeping balanced positions in cyclical rate-sensitive firms such as AMP (16.05%), BX (16.03%), and MS (15.90%). Real estate exposures (O at 8.69% and PLD at 1.81%) are sized more conservatively, reflecting their differentiated but secondary role within the broader financial ecosystem. By combining multiple optimization perspectives, the portfolio mirrors economic reality, where financial stocks respond differently to rate, credit, and liquidity regimes, while avoiding unintended concentration in a single macro-outcome.

Fit with the Macro Environment

The financial landscape is currently defined by conflicting signals regarding interest rate policy and credit cycle direction. Standard investment formulas struggle here because they often commit fully to a single outcome, either chasing past performance or betting entirely on a pivot that may not materialize. Our Ensemble approach fixes this by averaging signals across distinct frameworks (HRP, Black-Litterman, Mean-Variance), effectively diversifying our model risk rather than just our asset risk. This builds a portfolio resilient to multiple economic futures rather than one optimized for a single scenario. The result is a portfolio that earned over 17% by capturing the high beta of capital markets leaders while maintaining a defensive floor through structural diversification.

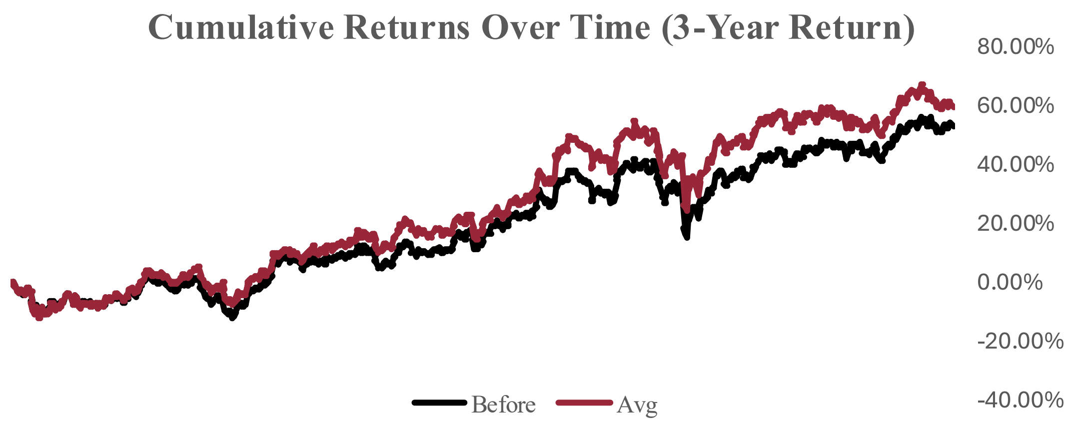
Practical Implications for Portfolio Construction

In practical terms, this framework improves position sizing by determining weights through a consensus lens before deploying capital. For example, where individual models either allocated 30% to V for safety or cut it to 0% to chase yield, the Ensemble model settles on a balanced 17.62% position. From a risk management standpoint, the framework prevents any single model's blind spot from dominating the portfolio, resulting in a high-conviction allocation to MKL (23.90%) while keeping a modest hedge in O (8.69%) to capture potential rate cuts without betting the house on them. This multi-vote approach grounds allocations in the overlap of competing philosophies, resulting in a Sharpe Ratio of 0.700 that balances historical efficiency with forward-looking adaptability.

Limitations and Caveats

However, there are important limitations to consider. The Ensemble model depends heavily on the assumption that the weaknesses of individual models will cancel each other out, meaning if all models share a common blind spot regarding systemic risk, the diversification benefit will be illusory. The model is also susceptible to signal dilution, where high-conviction insights from Black-Litterman are averaged down by backward-looking inputs, potentially leading to mediocrity if one specific view turns out to be correct. Finally, this approach requires maintaining four separate optimization engines simultaneously and assumes that the historical correlations driving the components will not suddenly invert during a liquidity crisis.

Avg Weightings	
AMP	16.05%
BX	16.03%
MKL	23.90%
MS	15.90%
O	8.69%
PLD	1.81%
V	17.62%



Performance	Annual Return	Annual Vol	Sharpe Ratio	Sortino Ratio
Avg	17.11%	18.03%	0.700	1.015